

Econ 695 – Midterm 1 (Reflection and Rewrite)

October 7, 2025

Date and Time: 8:00–9:15 AM, October 8th, 2025

Instructions: Write legibly and explain carefully what you are doing. Partial credit depends on clarity.

Question 1 (25 points ; 15 minutes)

Consider the distribution of individual incomes Y_i in a population that is divided into two groups, indicated by a random variable $D_i \in \{0, 1\}$. The population regression of interest is

$$Y_i = \beta_0 + \beta_1 D_i + u_i, \quad E[u_i] = E[u_i D_i] = 0,$$

and $\Pr(D_i = 1) = p$.

- (a) (10 points) Find $E[Y_i]$, the population mean of Y_i , as a function of β_0 and β_1 .

Initial thoughts:

Revised solution:

- (b) (15 points) Suppose $D_i = 1$ if person i is female and 0 otherwise. Also define $X_i = 1$ if i has at least a college degree, 0 otherwise. Find the conditional expectation function $E[Y_i \mid D_i, X_i]$. Propose a population regression that coincides with this CEF and explain.

Hint: carefully define the regressors based on binary D_i and X_i .

Initial thoughts:

Revised solution:

Question 2 (35 points ; 25 minutes)

Consider the following long and short regressions in the population:

$$\text{(Long)} \quad Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i,$$

$$\text{(Short)} \quad Y_i = \alpha_0 + \alpha_1 X_{1i} + \alpha_2 X_{2i} + v_i,$$

where $(\beta_0, \beta_1, \beta_2, \beta_3)$ in the long regression solve

$$\min_{\beta_0, \beta_1, \beta_2, \beta_3} E[(Y_i - \beta_0 - \beta_1 X_{1i} - \beta_2 X_{2i} - \beta_3 X_{3i})^2],$$

and $(\alpha_0, \alpha_1, \alpha_2)$ in the short regression solve

$$\min_{\alpha_0, \alpha_1, \alpha_2} E[(Y_i - \alpha_0 - \alpha_1 X_{1i} - \alpha_2 X_{2i})^2].$$

(a) (10 points) Derive a formula relating α_1 and β_1, β_3 .

(b) (10 points) Suppose $\text{cov}(X_{1i}, X_{3i}) = 0$ and $\text{cov}(X_{2i}, X_{3i}) = 0$. Explain.

(c) (15 points) Suppose $\text{cov}(X_{2i}, X_{3i}) = 0$ but $\text{cov}(X_{1i}, X_{3i}) \neq 0$. Does $\alpha_2 = \beta_2$? Explain.

Initial approach:

Revised reasoning:

Question 3 (40 points ; 30 minutes)

In Lecture 3b we examined the role of education in driving the wage gap between natives and immigrants. We considered four OLS regressions:

- (1) $Y_i = \pi_0 + \pi_1 \text{Imm}_i$,
- (2) $Y_i = \beta_0 + \beta_1 \text{Imm}_i + \beta_2 \text{Educ}_i$,
- (3) $Y_i = \theta_0^a + \theta_1^a \text{Educ}_i$ (among natives),
- (4) $Y_i = \theta_0^b + \theta_1^b \text{Educ}_i$ (among immigrants).

Y_i is log wage, $\text{Imm}_i = 1$ if immigrant, and Educ_i = years of schooling. Models (1)–(2) use the pooled sample (natives + immigrants); (3)–(4) are separate by group. The table provides estimated coefficients:

	(1)	(2)	(3) Natives	(4) Immigrants
Constant	3.013	1.546	3.013	1.676
Immigrant	−0.247	−0.072	—	—
Education (yrs)	—	0.104	0.117	0.088
MSE	0.757	0.695	0.689	0.707
Adj. R^2	0.018	0.173	0.146	0.208
Sample Size	19,092	19,092	14,921	4,141

- (a) (10 points) Compute the average log-wage gap $g_b - g_a$ between immigrants and natives. Provide a numeric answer and explain.

(b) (15 points) On average immigrants have 12 years of schooling, natives 14. How much of the gap in (a) is explained by education differences? Use either the pooled regression (2) or the separate models (3)–(4).

(c) (15 points) The pooled return to education β_2 in (2) is a weighted average of returns among natives and immigrants:

$$\beta_2 = w_a \theta_1^a + w_b \theta_1^b.$$

By Frisch–Waugh–Lovell, define the residuals $\ell_i = \text{Educ}_i - \pi_0 - \pi_1 \text{Imm}_i$. Let $p_a = N_a/N$ be the native share of the sample, $\text{var}(\ell_i)$ the sample variance of ℓ_i , and $\text{var}(x_i \mid a)$, $\text{var}(x_i \mid b)$ the education variances within natives and immigrants.

Find w_a and w_b as functions of p_a , $\text{var}(\ell_i)$, and the group-specific variances.

Initial reasoning:

Updated solution: